

LOSO (glove_ml_explorer) Experiment SC_left

Static · full sensors, LEFT hand only, no Nothing
 Generated 2026-05-12 00:30:56

Experiment configuration

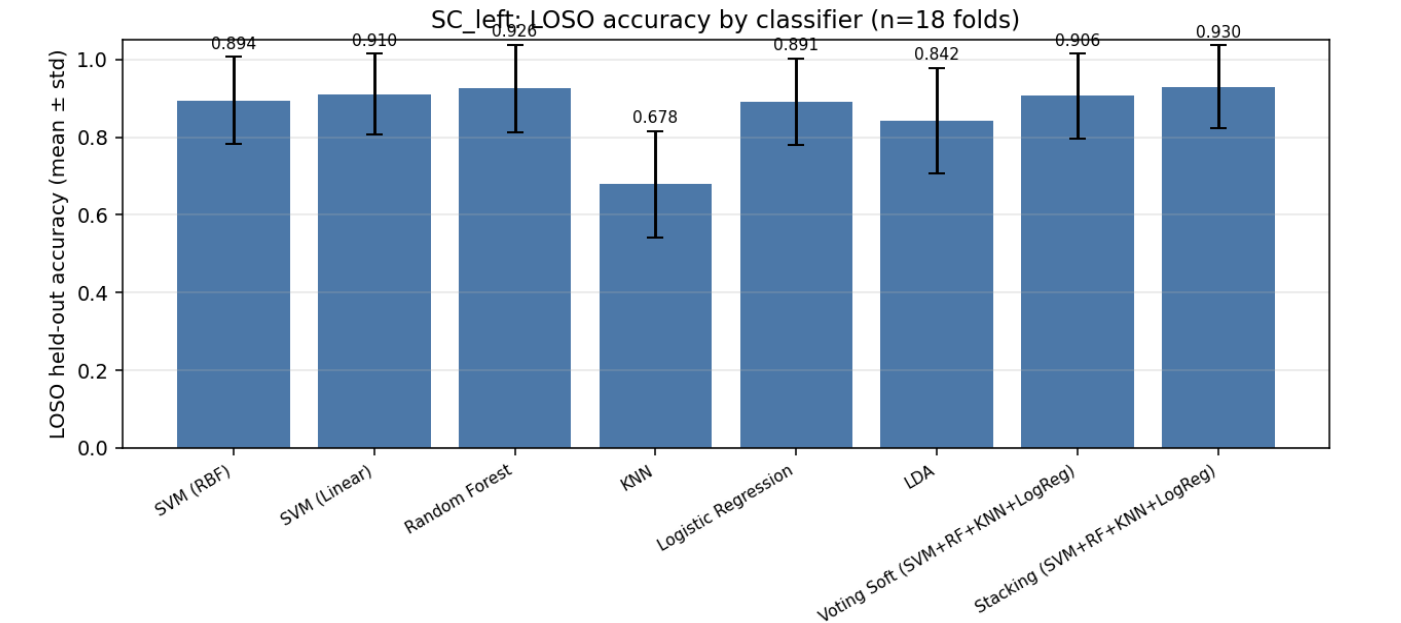
Class set	6 classes excluding Double_Nothing
Classes	Double_Closed_Fist, Double_Okay, Double_Open_Palm, Double_Phone, Double_Pistol, Double_Spiderman
Sensor modalities	flex, YPR, accel
Hands	left
Segments	thumb, index, middle, ring, pinky, wrist
Sensor channels	76
Sequence length	90 steps (resampled)
Filter	off
Feature mode	stats+fft
Normalisation	minmax
Augmentation	off
Inner CV folds	5
LOSO subjects	18: 10_Josh, 11_Mansh, 12_Marcus, 13_Nat, 14_StephenV2, 16_Bella, 17_Raquel, 1_Alan, 2_Alex, 3_Anghad, 4_Daniel, 5_Georgia, 6_Harry, 7_Henry, 8_Jestin, 9_Joselyn, Stephen, Val

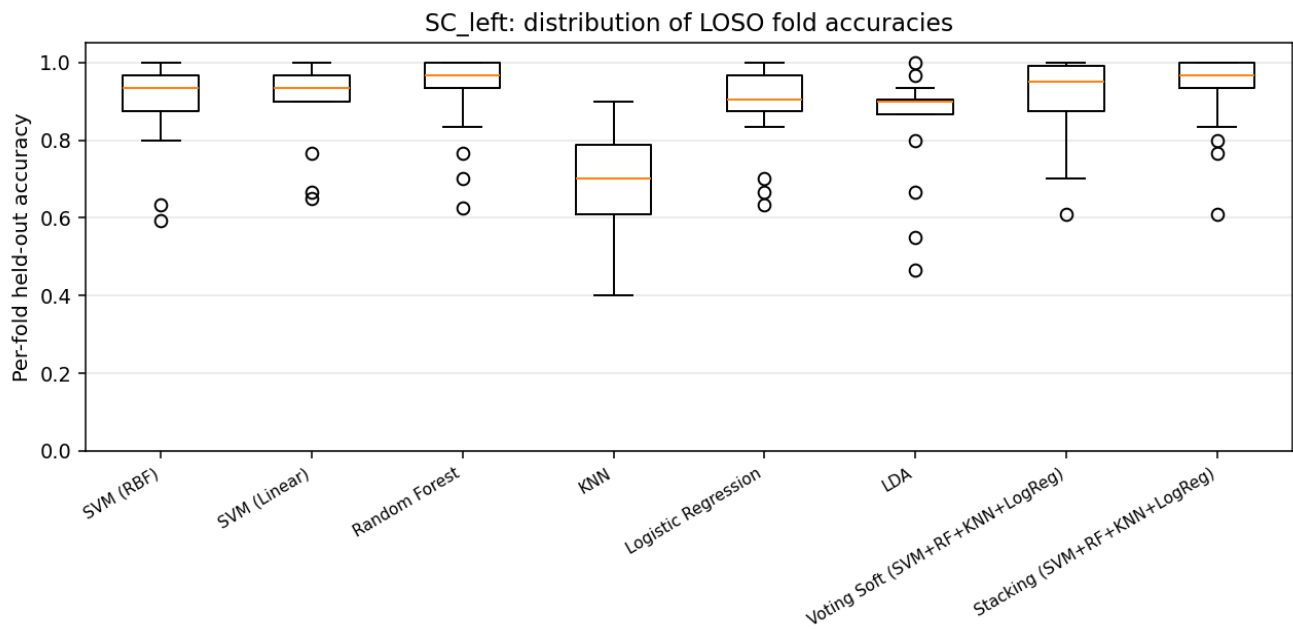
LOSO summary across all subjects

Classifier	Params*	LOSO mean	LOSO std	Min fold	Max fold	Inner CV mean
SVM (RBF)	—	0.8944	0.1131	0.5917	1.0000	0.9308
SVM (Linear)	61,575	0.9105	0.1042	0.6500	1.0000	0.9366
Random Forest	18,836	0.9255	0.1121	0.6250	1.0000	0.9440
KNN	—	0.6784	0.1375	0.4000	0.9000	0.7305
Logistic Regression	24,630	0.8911	0.1116	0.6333	1.0000	0.9358
LDA	24,630	0.8424	0.1364	0.4667	1.0000	0.8907
Voting Soft (SVM+RF+KNN+LogReg)	—	0.9061	0.1095	0.6083	1.0000	0.9470
Stacking (SVM+RF+KNN+LogReg)	—	0.9302	0.1057	0.6083	1.0000	0.9604

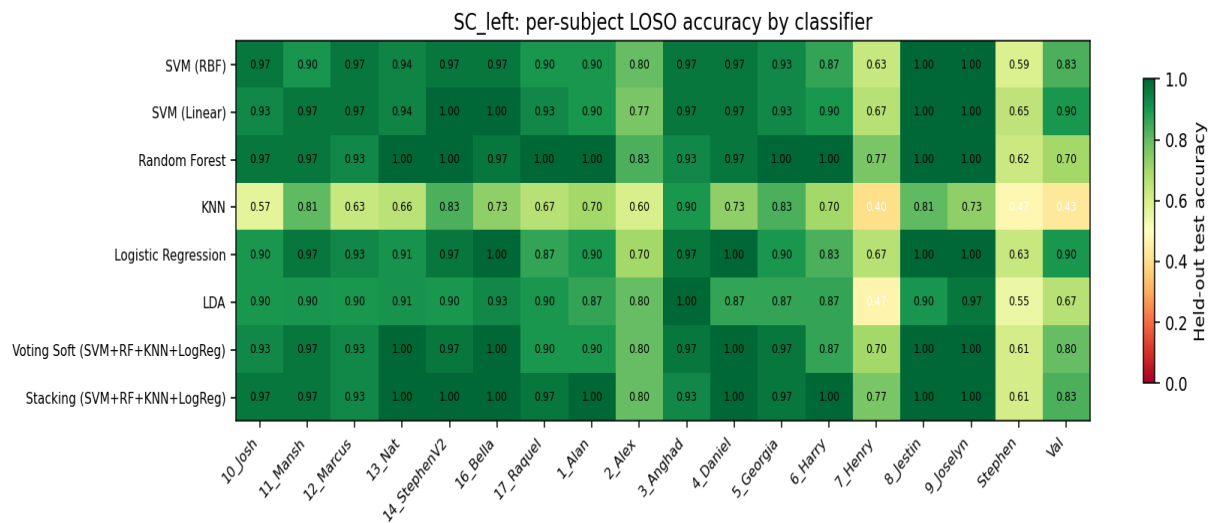
*Params: linear models use coef_+intercept size; tree ensembles use total node count across trees. SVM-RBF / KNN / Voting / Stacking are non-parametric or composite, so they show as —.

Best classifier by LOSO mean: **Stacking (SVM+RF+KNN+LogReg)**





Per-subject LOSO accuracy heatmap



Per-fold detail (held-out subject = each row)

Held-out subj	Train trials	SVM (RBF)	SVM (Linear)	Random Forest	KNN	Logistic Regression	LDA	Voting Soft (SVM+RF+KNN+LogReg)	Stacking (SVM+RF+KNN+LogReg)
10_Josh	604	0.967	0.933	0.967	0.567	0.900	0.900	0.933	0.967
11_Mansh	603	0.903	0.968	0.968	0.806	0.968	0.903	0.968	0.968
12_Marcus	604	0.967	0.967	0.933	0.633	0.933	0.900	0.933	0.933
13_Nat	602	0.938	0.938	1.000	0.656	0.906	0.906	1.000	1.000
14_StephenV2	604	0.967	1.000	1.000	0.833	0.967	0.900	0.967	1.000
16_Bella	604	0.967	1.000	0.967	0.733	1.000	0.933	1.000	1.000
17_Raquel	604	0.900	0.933	1.000	0.667	0.867	0.900	0.900	0.967
1_Alan	604	0.900	0.900	1.000	0.700	0.900	0.867	0.900	1.000
2_Alex	604	0.800	0.767	0.833	0.600	0.700	0.800	0.800	0.800
3_Anghad	604	0.967	0.967	0.933	0.900	0.967	1.000	0.967	0.933
4_Daniel	604	0.967	0.967	0.967	0.733	1.000	0.867	1.000	1.000
5_Georgia	604	0.933	0.933	1.000	0.833	0.900	0.867	0.967	0.967
6_Harry	604	0.867	0.900	1.000	0.700	0.833	0.867	0.867	1.000
7_Henry	604	0.633	0.667	0.767	0.400	0.667	0.467	0.700	0.767
8_Jestin	603	1.000	1.000	1.000	0.806	1.000	0.903	1.000	1.000
9_Joselyn	604	1.000	1.000	1.000	0.733	1.000	0.967	1.000	1.000
Stephen	514	0.592	0.650	0.625	0.475	0.633	0.550	0.608	0.608
Val	604	0.833	0.900	0.700	0.433	0.900	0.667	0.800	0.833

Classifier: SVM (RBF)

Params (best-effort): —

LOSO mean accuracy: 0.8944 ± 0.1131 (min 0.5917, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	5/5	5/5	5/5	4/5	5/5	5/5	0.967
11_Mansh	3/6	5/5	5/5	5/5	5/5	5/5	0.903
12_Marcus	5/5	5/5	5/5	4/5	5/5	5/5	0.967
13_Nat	6/6	3/5	5/5	5/5	6/6	5/5	0.938
14_StephenV2	5/5	5/5	4/5	5/5	5/5	5/5	0.967
16_Bella	5/5	5/5	5/5	4/5	5/5	5/5	0.967
17_Raquel	5/5	2/5	5/5	5/5	5/5	5/5	0.900
1_Alan	3/5	5/5	5/5	4/5	5/5	5/5	0.900
2_Alex	4/5	5/5	5/5	0/5	5/5	5/5	0.800
3_Anghad	4/5	5/5	5/5	5/5	5/5	5/5	0.967
4_Daniel	5/5	5/5	5/5	4/5	5/5	5/5	0.967
5_Georgia	5/5	4/5	4/5	5/5	5/5	5/5	0.933
6_Harry	5/5	4/5	5/5	2/5	5/5	5/5	0.867
7_Henry	5/5	4/5	1/5	4/5	0/5	5/5	0.633
8_Jestin	6/6	5/5	5/5	5/5	5/5	5/5	1.000
9_Joselyn	5/5	5/5	5/5	5/5	5/5	5/5	1.000
Stephen	10/20	9/20	14/20	8/20	14/20	16/20	0.592
Val	3/5	5/5	4/5	3/5	5/5	5/5	0.833
MEAN	0.878	0.881	0.906	0.789	0.928	0.989	0.894

Classifier: SVM (Linear)

Params (best-effort): 61,575

LOSO mean accuracy: 0.9105 ± 0.1042 (min 0.6500, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	5/5	5/5	5/5	3/5	5/5	5/5	0.933
11_Mansh	5/6	5/5	5/5	5/5	5/5	5/5	0.968
12_Marcus	5/5	5/5	5/5	4/5	5/5	5/5	0.967
13_Nat	6/6	4/5	4/5	5/5	6/6	5/5	0.938
14_StephenV2	5/5	5/5	5/5	5/5	5/5	5/5	1.000
16_Bella	5/5	5/5	5/5	5/5	5/5	5/5	1.000
17_Raquel	5/5	3/5	5/5	5/5	5/5	5/5	0.933
1_Alan	3/5	5/5	5/5	4/5	5/5	5/5	0.900
2_Alex	3/5	5/5	5/5	0/5	5/5	5/5	0.767
3_Anghad	4/5	5/5	5/5	5/5	5/5	5/5	0.967
4_Daniel	4/5	5/5	5/5	5/5	5/5	5/5	0.967
5_Georgia	5/5	4/5	4/5	5/5	5/5	5/5	0.933
6_Harry	5/5	5/5	5/5	2/5	5/5	5/5	0.900
7_Henry	5/5	5/5	1/5	4/5	1/5	4/5	0.667
8_Jestin	6/6	5/5	5/5	5/5	5/5	5/5	1.000
9_Joselyn	5/5	5/5	5/5	5/5	5/5	5/5	1.000
Stephen	10/20	9/20	16/20	13/20	14/20	16/20	0.650
Val	3/5	5/5	5/5	4/5	5/5	5/5	0.900
MEAN	0.874	0.925	0.922	0.825	0.939	0.978	0.910

Classifier: Random Forest

Params (best-effort): 18,836

LOSO mean accuracy: 0.9255 ± 0.1121 (min 0.6250, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	5/5	5/5	5/5	4/5	5/5	5/5	0.967
11_Mansh	6/6	4/5	5/5	5/5	5/5	5/5	0.968
12_Marcus	5/5	5/5	5/5	3/5	5/5	5/5	0.933
13_Nat	6/6	5/5	5/5	5/5	6/6	5/5	1.000
14_StephenV2	5/5	5/5	5/5	5/5	5/5	5/5	1.000
16_Bella	5/5	4/5	5/5	5/5	5/5	5/5	0.967
17_Raquel	5/5	5/5	5/5	5/5	5/5	5/5	1.000
1_Alan	5/5	5/5	5/5	5/5	5/5	5/5	1.000
2_Alex	5/5	5/5	5/5	0/5	5/5	5/5	0.833
3_Anghad	3/5	5/5	5/5	5/5	5/5	5/5	0.933
4_Daniel	4/5	5/5	5/5	5/5	5/5	5/5	0.967
5_Georgia	5/5	5/5	5/5	5/5	5/5	5/5	1.000
6_Harry	5/5	5/5	5/5	5/5	5/5	5/5	1.000
7_Henry	5/5	5/5	4/5	4/5	0/5	5/5	0.767
8_Jestin	6/6	5/5	5/5	5/5	5/5	5/5	1.000
9_Joselyn	5/5	5/5	5/5	5/5	5/5	5/5	1.000
Stephen	5/20	11/20	19/20	5/20	17/20	18/20	0.625
Val	3/5	5/5	3/5	2/5	5/5	3/5	0.700
MEAN	0.903	0.953	0.964	0.825	0.936	0.972	0.926

Classifier: KNN

Params (best-effort): —

LOSO mean accuracy: 0.6784 ± 0.1375 (min 0.4000, max 0.9000)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	4/5	1/5	3/5	0/5	4/5	5/5	0.567
11_Mansh	4/6	5/5	5/5	1/5	5/5	5/5	0.806
12_Marcus	3/5	2/5	4/5	0/5	5/5	5/5	0.633
13_Nat	6/6	2/5	2/5	0/5	6/6	5/5	0.656
14_StephenV2	4/5	5/5	4/5	2/5	5/5	5/5	0.833
16_Bella	3/5	4/5	4/5	1/5	5/5	5/5	0.733
17_Raquel	4/5	0/5	5/5	1/5	5/5	5/5	0.667
1_Alan	4/5	2/5	4/5	1/5	5/5	5/5	0.700
2_Alex	5/5	2/5	1/5	1/5	4/5	5/5	0.600
3_Anghad	4/5	4/5	4/5	5/5	5/5	5/5	0.900
4_Daniel	4/5	4/5	4/5	0/5	5/5	5/5	0.733
5_Georgia	5/5	3/5	5/5	2/5	5/5	5/5	0.833
6_Harry	5/5	4/5	2/5	0/5	5/5	5/5	0.700
7_Henry	2/5	4/5	2/5	0/5	0/5	4/5	0.400
8_Jestin	5/6	5/5	4/5	1/5	5/5	5/5	0.806
9_Joselyn	3/5	4/5	5/5	0/5	5/5	5/5	0.733
Stephen	6/20	10/20	6/20	3/20	15/20	17/20	0.475
Val	3/5	1/5	4/5	0/5	3/5	2/5	0.433
MEAN	0.744	0.606	0.706	0.175	0.886	0.947	0.678

Classifier: Logistic Regression

Params (best-effort): 24,630

LOSO mean accuracy: 0.8911 ± 0.1116 (min 0.6333, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	5/5	4/5	5/5	3/5	5/5	5/5	0.900
11_Mansh	5/6	5/5	5/5	5/5	5/5	5/5	0.968
12_Marcus	5/5	5/5	5/5	3/5	5/5	5/5	0.933
13_Nat	6/6	5/5	2/5	5/5	6/6	5/5	0.906
14_StephenV2	5/5	5/5	5/5	5/5	4/5	5/5	0.967
16_Bella	5/5	5/5	5/5	5/5	5/5	5/5	1.000
17_Raquel	5/5	1/5	5/5	5/5	5/5	5/5	0.867
1_Alan	3/5	5/5	5/5	4/5	5/5	5/5	0.900
2_Alex	2/5	5/5	5/5	0/5	4/5	5/5	0.700
3_Anghad	4/5	5/5	5/5	5/5	5/5	5/5	0.967
4_Daniel	5/5	5/5	5/5	5/5	5/5	5/5	1.000
5_Georgia	5/5	4/5	5/5	3/5	5/5	5/5	0.900
6_Harry	5/5	4/5	5/5	1/5	5/5	5/5	0.833
7_Henry	5/5	5/5	1/5	4/5	0/5	5/5	0.667
8_Jestin	6/6	5/5	5/5	5/5	5/5	5/5	1.000
9_Joselyn	5/5	5/5	5/5	5/5	5/5	5/5	1.000
Stephen	7/20	9/20	16/20	12/20	15/20	17/20	0.633
Val	3/5	5/5	5/5	4/5	5/5	5/5	0.900
MEAN	0.866	0.892	0.911	0.778	0.908	0.992	0.891

Classifier: LDA

Params (best-effort): 24,630

LOSO mean accuracy: 0.8424 ± 0.1364 (min 0.4667, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	4/5	4/5	5/5	4/5	5/5	5/5	0.900
11_Mansh	3/6	5/5	5/5	5/5	5/5	5/5	0.903
12_Marcus	5/5	5/5	3/5	4/5	5/5	5/5	0.900
13_Nat	5/6	4/5	4/5	5/5	6/6	5/5	0.906
14_StephenV2	5/5	5/5	4/5	3/5	5/5	5/5	0.900
16_Bella	5/5	5/5	5/5	4/5	4/5	5/5	0.933
17_Raquel	5/5	5/5	4/5	3/5	5/5	5/5	0.900
1_Alan	3/5	5/5	5/5	3/5	5/5	5/5	0.867
2_Alex	3/5	5/5	5/5	1/5	5/5	5/5	0.800
3_Anghad	5/5	5/5	5/5	5/5	5/5	5/5	1.000
4_Daniel	5/5	5/5	5/5	1/5	5/5	5/5	0.867
5_Georgia	3/5	4/5	4/5	5/5	5/5	5/5	0.867
6_Harry	5/5	5/5	5/5	1/5	5/5	5/5	0.867
7_Henry	1/5	5/5	1/5	4/5	1/5	2/5	0.467
8_Jestin	5/6	5/5	4/5	4/5	5/5	5/5	0.903
9_Joselyn	4/5	5/5	5/5	5/5	5/5	5/5	0.967
Stephen	1/20	6/20	19/20	9/20	14/20	17/20	0.550
Val	1/5	4/5	4/5	3/5	5/5	3/5	0.667
MEAN	0.723	0.917	0.864	0.692	0.928	0.936	0.842

Classifier: Voting Soft (SVM+RF+KNN+LogReg)

Params (best-effort): —

LOSO mean accuracy: 0.9061 ± 0.1095 (min 0.6083, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	5/5	5/5	5/5	3/5	5/5	5/5	0.933
11_Mansh	5/6	5/5	5/5	5/5	5/5	5/5	0.968
12_Marcus	5/5	5/5	5/5	3/5	5/5	5/5	0.933
13_Nat	6/6	5/5	5/5	5/5	6/6	5/5	1.000
14_StephenV2	5/5	5/5	4/5	5/5	5/5	5/5	0.967
16_Bella	5/5	5/5	5/5	5/5	5/5	5/5	1.000
17_Raquel	5/5	2/5	5/5	5/5	5/5	5/5	0.900
1_Alan	3/5	5/5	5/5	4/5	5/5	5/5	0.900
2_Alex	4/5	5/5	5/5	0/5	5/5	5/5	0.800
3_Anghad	4/5	5/5	5/5	5/5	5/5	5/5	0.967
4_Daniel	5/5	5/5	5/5	5/5	5/5	5/5	1.000
5_Georgia	5/5	4/5	5/5	5/5	5/5	5/5	0.967
6_Harry	5/5	4/5	5/5	2/5	5/5	5/5	0.867
7_Henry	5/5	5/5	2/5	4/5	0/5	5/5	0.700
8_Jestin	6/6	5/5	5/5	5/5	5/5	5/5	1.000
9_Joselyn	5/5	5/5	5/5	5/5	5/5	5/5	1.000
Stephen	7/20	9/20	16/20	9/20	15/20	17/20	0.608
Val	3/5	5/5	4/5	3/5	5/5	4/5	0.800
MEAN	0.888	0.914	0.933	0.792	0.931	0.981	0.906

Classifier: Stacking (SVM+RF+KNN+LogReg)

Params (best-effort): —

LOSO mean accuracy: 0.9302 ± 0.1057 (min 0.6083, max 1.0000)

Per-fold per-class held-out accuracy

Held-out	Double_Closed_Fist	Double_Okay	Double_Open_Palm	Double_Phone	Double_Pistol	Double_Spiderman	Overall
10_Josh	5/5	5/5	5/5	4/5	5/5	5/5	0.967
11_Mansh	5/6	5/5	5/5	5/5	5/5	5/5	0.968
12_Marcus	5/5	5/5	5/5	3/5	5/5	5/5	0.933
13_Nat	6/6	5/5	5/5	5/5	6/6	5/5	1.000
14_StephenV2	5/5	5/5	5/5	5/5	5/5	5/5	1.000
16_Bella	5/5	5/5	5/5	5/5	5/5	5/5	1.000
17_Raquel	5/5	4/5	5/5	5/5	5/5	5/5	0.967
1_Alan	5/5	5/5	5/5	5/5	5/5	5/5	1.000
2_Alex	4/5	5/5	5/5	0/5	5/5	5/5	0.800
3_Anghad	3/5	5/5	5/5	5/5	5/5	5/5	0.933
4_Daniel	5/5	5/5	5/5	5/5	5/5	5/5	1.000
5_Georgia	5/5	4/5	5/5	5/5	5/5	5/5	0.967
6_Harry	5/5	5/5	5/5	5/5	5/5	5/5	1.000
7_Henry	5/5	5/5	3/5	5/5	0/5	5/5	0.767
8_Jestin	6/6	5/5	5/5	5/5	5/5	5/5	1.000
9_Joselyn	5/5	5/5	5/5	5/5	5/5	5/5	1.000
Stephen	5/20	9/20	16/20	11/20	15/20	17/20	0.608
Val	3/5	5/5	5/5	3/5	5/5	4/5	0.833
MEAN	0.894	0.947	0.967	0.864	0.931	0.981	0.930